

BPC Training

Module 2 – Overview of Process Monitoring & Statistical Process Control

9th – 11th Jan 2023

CC-SISPI FTKKP UMP

Topics

1. Process Monitoring Overview

- Process Data Variation
- Motivation & Goal of Process Monitoring
- Classification & Desirable Characteristics

2. Statistical Process Control (SPC)

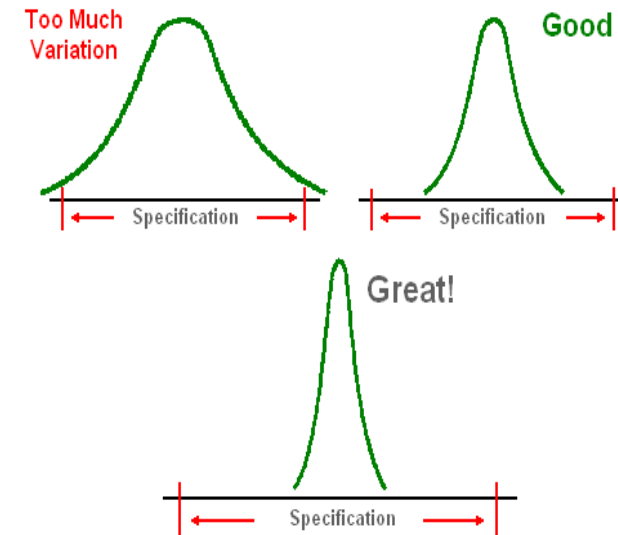
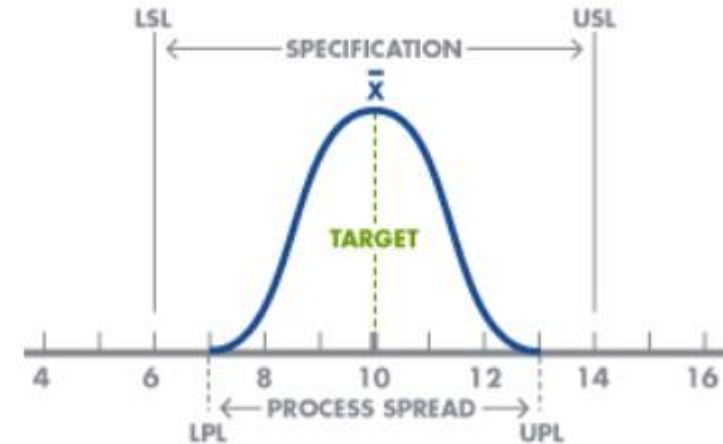
- Normal distribution model
- Development of control charts
- Interpretation of control charts

Process Monitoring Overview

Process Data Variation

The ultimate aim of any production systems, in the context of chemical and process industries, is to produce maximum amount of consistently high quality products as per requested and specified by the customers.

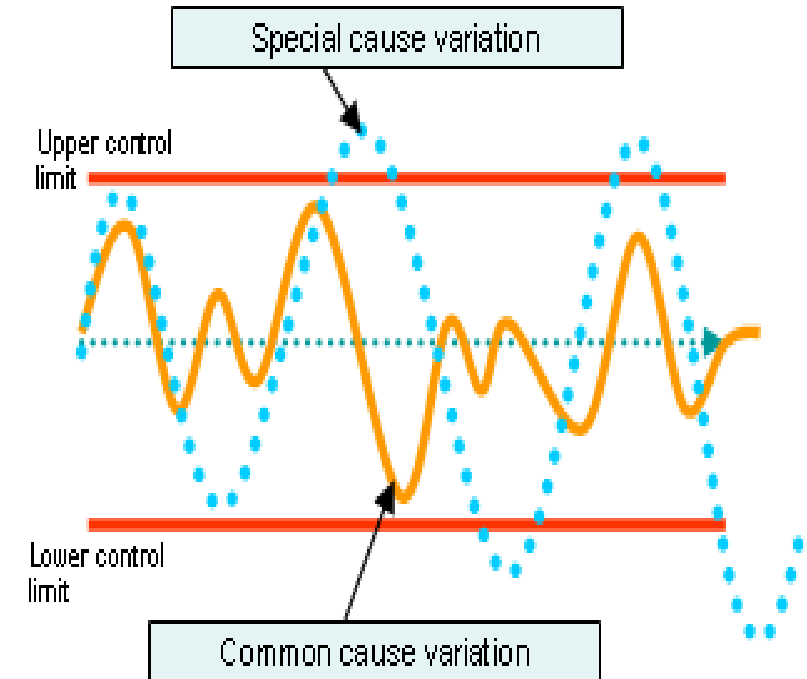
This is regarded as **highly challenging** due to the nature of the processes that **always change over time** and are also affected by various factors such as variations of raw materials as well as operating conditions, the presence of disturbances and also modification of the process technologies.



Process Monitoring Overview

Process Data Variation

- Normal variation is caused by the cumulative effects of a number of largely unavoidable phenomena such as electrical measurement noise, turbulence, and random fluctuations in feedstock or catalyst preparation.
- Abnormal variation (fault operation) is referred to *special cause* or an *assignable cause* such as disturbances, equipment degradation, upsets in process condition, changing in operating modes, signal/communication failure.



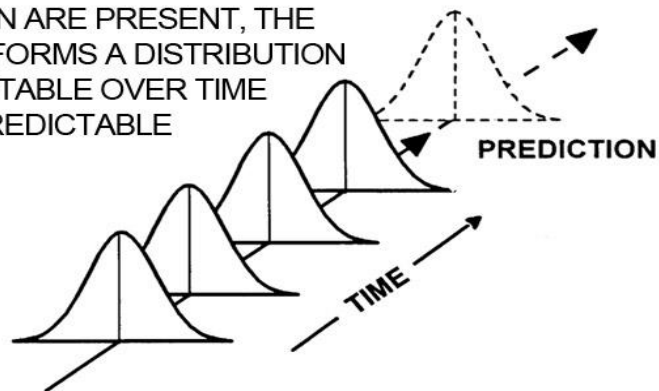
Process Monitoring Overview

Process Data Variation

In any of the situations, one of the main critical problems is to systematically manage and remove the occurrence of faulty or abnormal operating conditions in the routine process operation, and at the same time struggling **to be consistent upon the prime objective**.

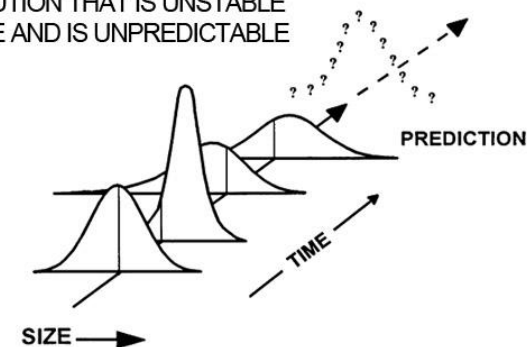
A Stable Process

IF ONLY COMMON CAUSES OF VARIATION ARE PRESENT, THE OUTPUT FORMS A DISTRIBUTION THAT IS STABLE OVER TIME AND IS PREDICTABLE



An Unstable Process

IF SPECIAL CAUSES OF VARIATION ARE PRESENT, THE OUTPUT FORMS A DISTRIBUTION THAT IS UNSTABLE OVER TIME AND IS UNPREDICTABLE



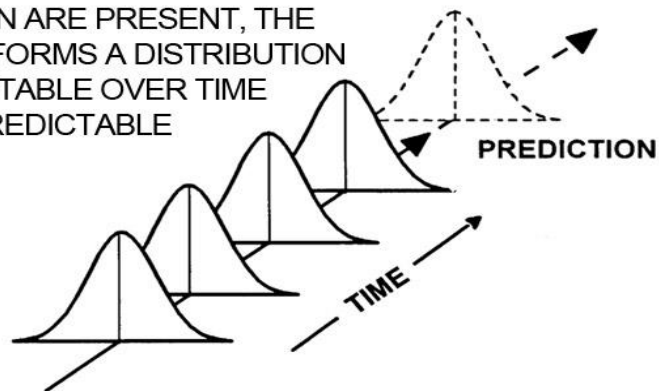
Process Monitoring Overview

Process Data Variation

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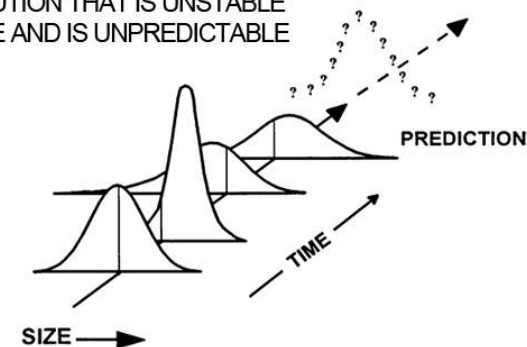
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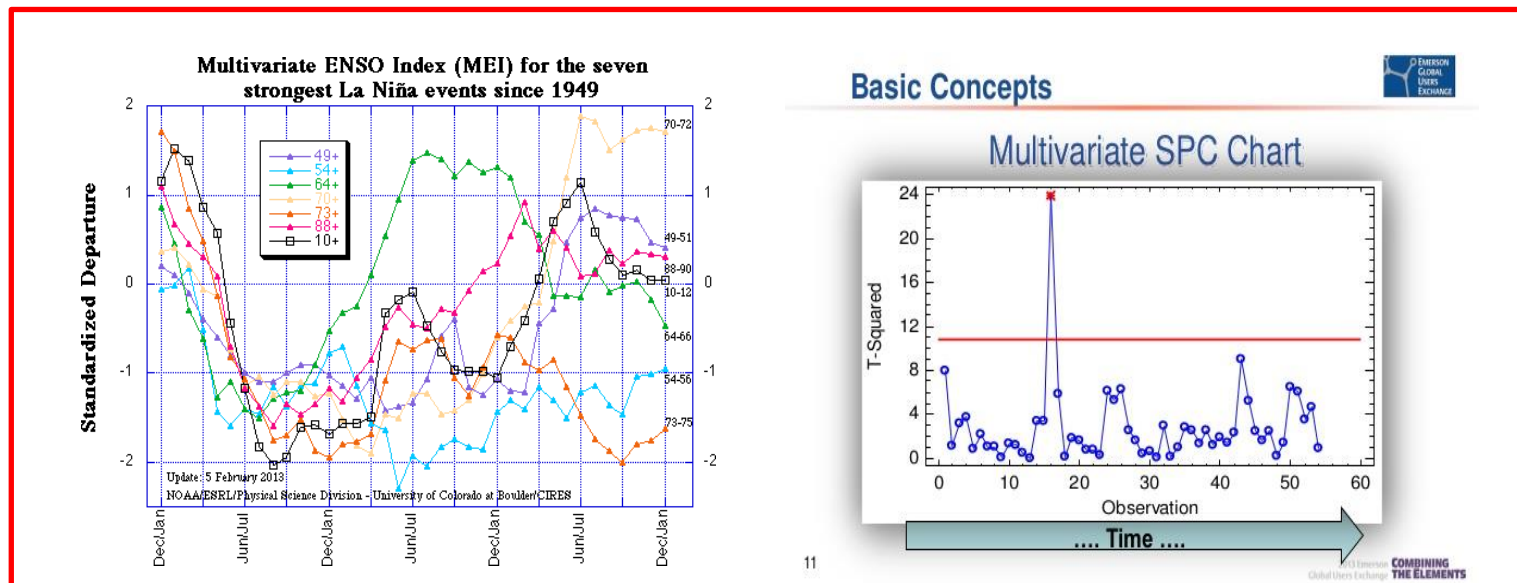
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Process Monitoring Overview

Motivation & Goal of Process Monitoring

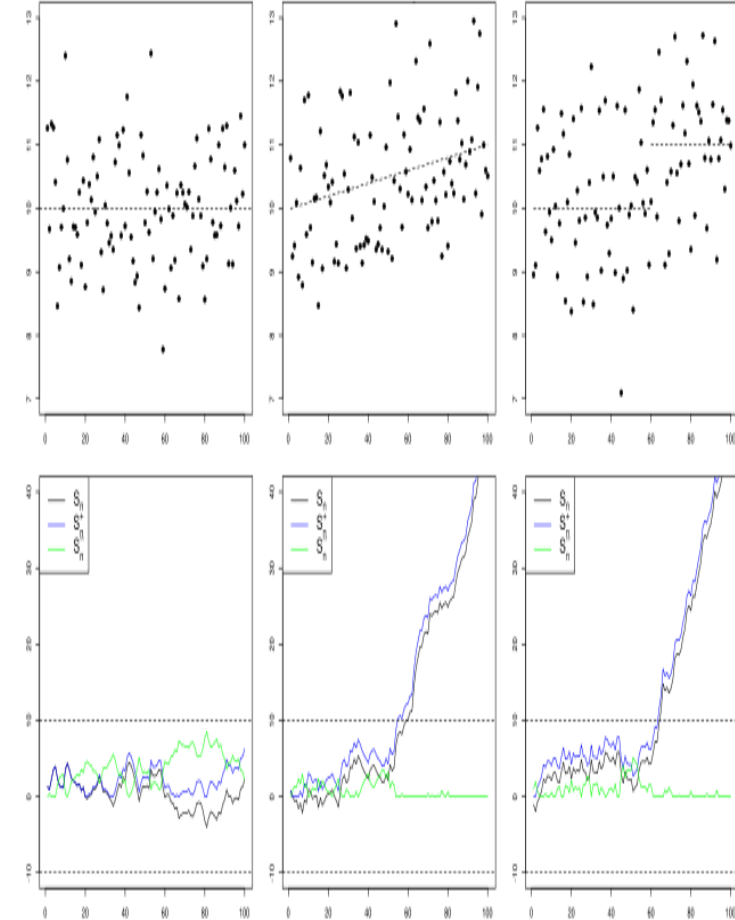
By definition, **PROCESS MONITORING** is a systematic mechanism in order to identify the most important process variables which define the overall process quality as well as to ensure that those quality variables are operated under the pre-specified 'normal' circumstances.



Process Monitoring Overview

Motivation & Goal of Process Monitoring

The primary aim/goal of process monitoring: **to provide early warning** by way of identification of possible faults, malfunctions or disturbances that might occur in the process, particularly in order to avoid any dangerous or catastrophic situation.

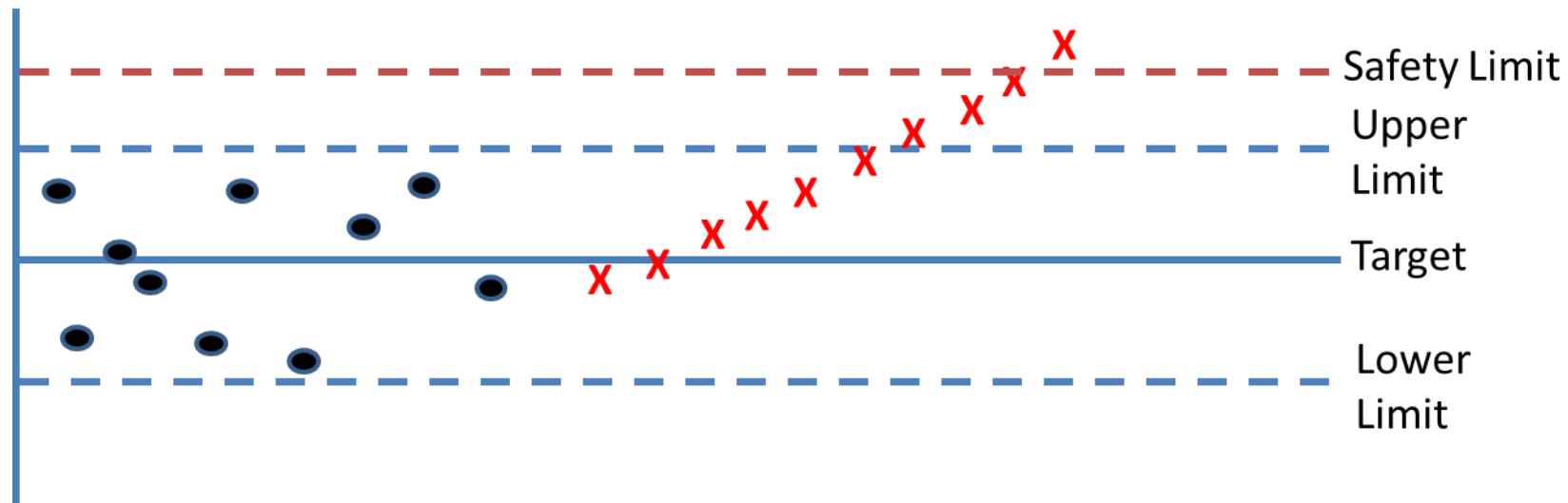


Process Monitoring Overview

Motivation & Goal of Process Monitoring

The general objectives/benefits of process monitoring are:

- 1. Routine Monitoring.** Ensure that process variables are within the specified limits.
- 2. Detection and Diagnosis.** Detect abnormal process operation and diagnose the root cause.
- 3. Preventive Monitoring.** Detect abnormal situations early enough so that corrective action can be taken before the process is seriously upset.



Process Monitoring Overview

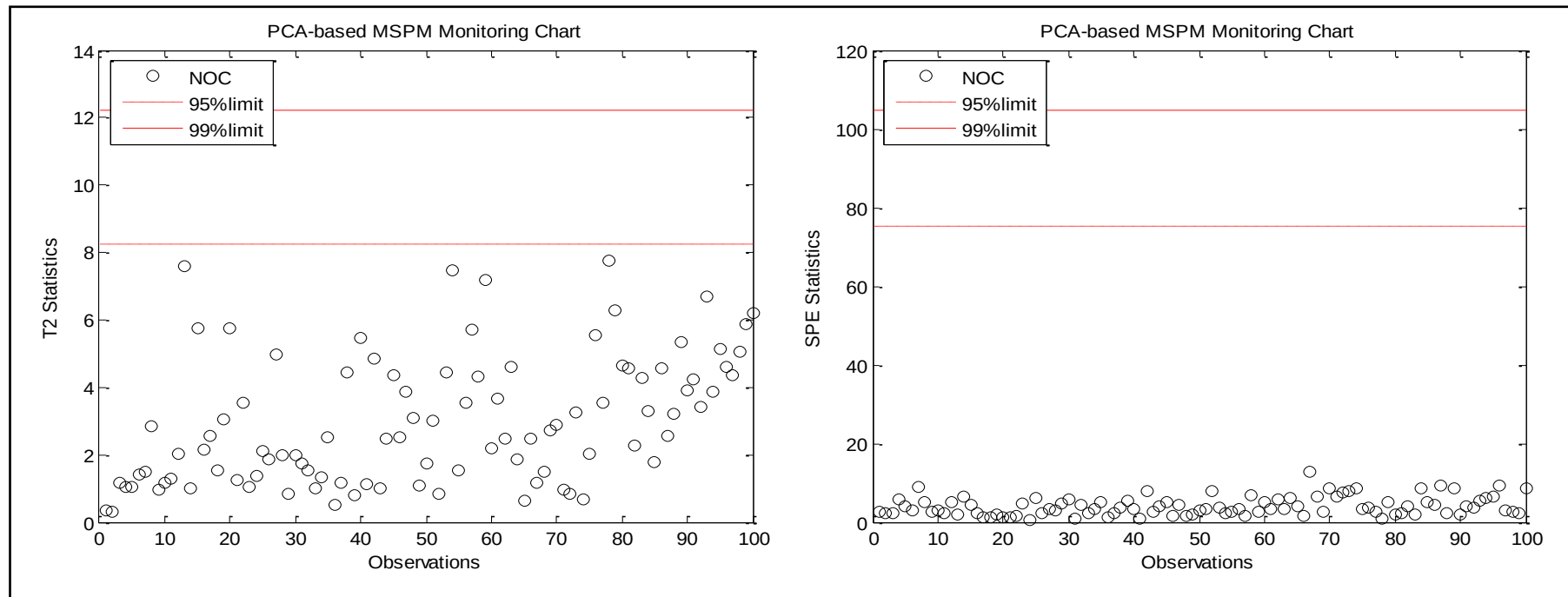
Motivation & Goal of Process Monitoring

- Two types of monitoring methods:
 1. Univariate Monitoring => Statistical Process Control (SPC).
 2. Multivariate Monitoring => Multivariate Statistical Process Control/Monitoring (MSPC/MSPM)
- Two phases of building a monitoring application:
 - Phase I: **NOC model development by means of control limits.**
To gain an understanding of the process and to establish a statistical benchmark for the likely future process outcomes -> low false alarm rate/analysis.
 - Phase II: **Fault detection and identification operation.**
Observing the process in real time by comparing the new process data with the pre-specified model that established during the first phase -> normal (progression within limits) /abnormal (consistent limits violation happened)!

Process Monitoring Overview

Motivation & Goal of Process Monitoring

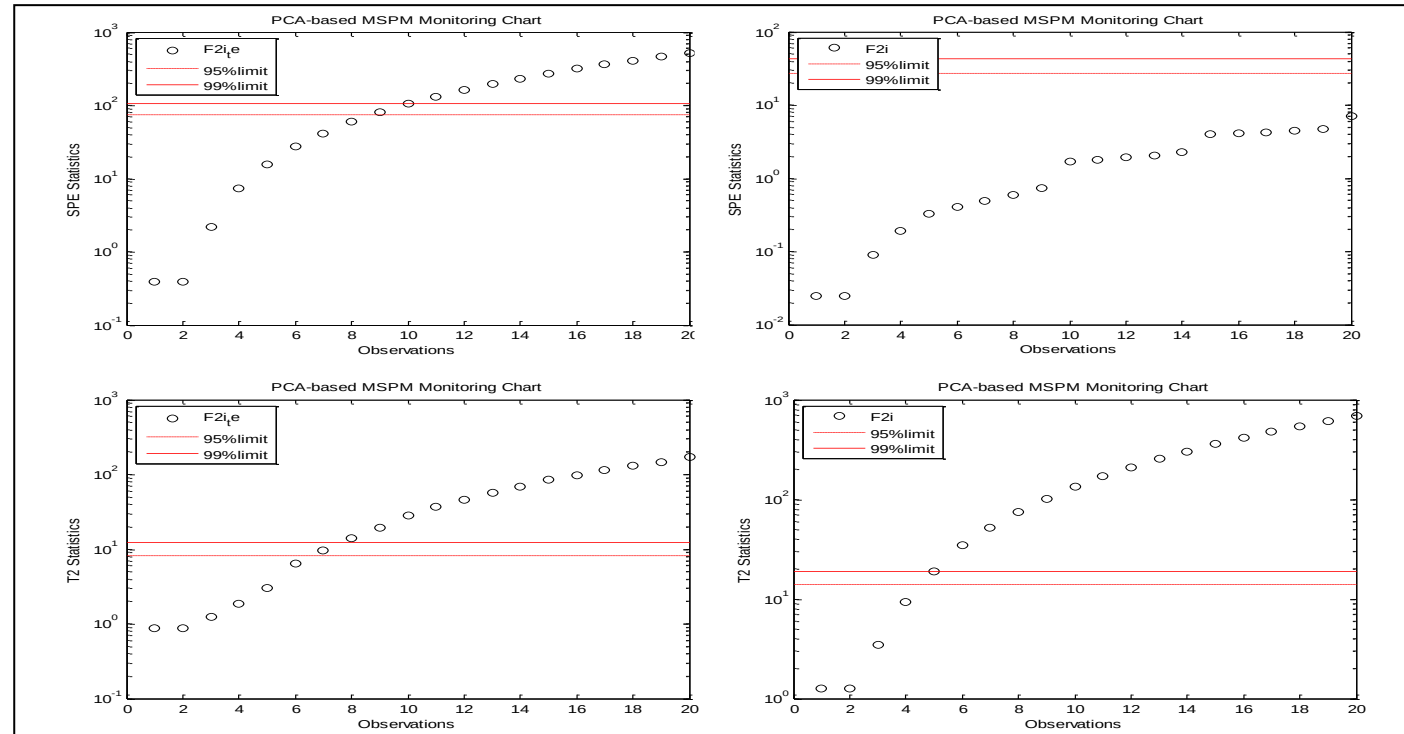
PHASE I: NOC Model Development



Process Monitoring Overview

Motivation & Goal of Process Monitoring

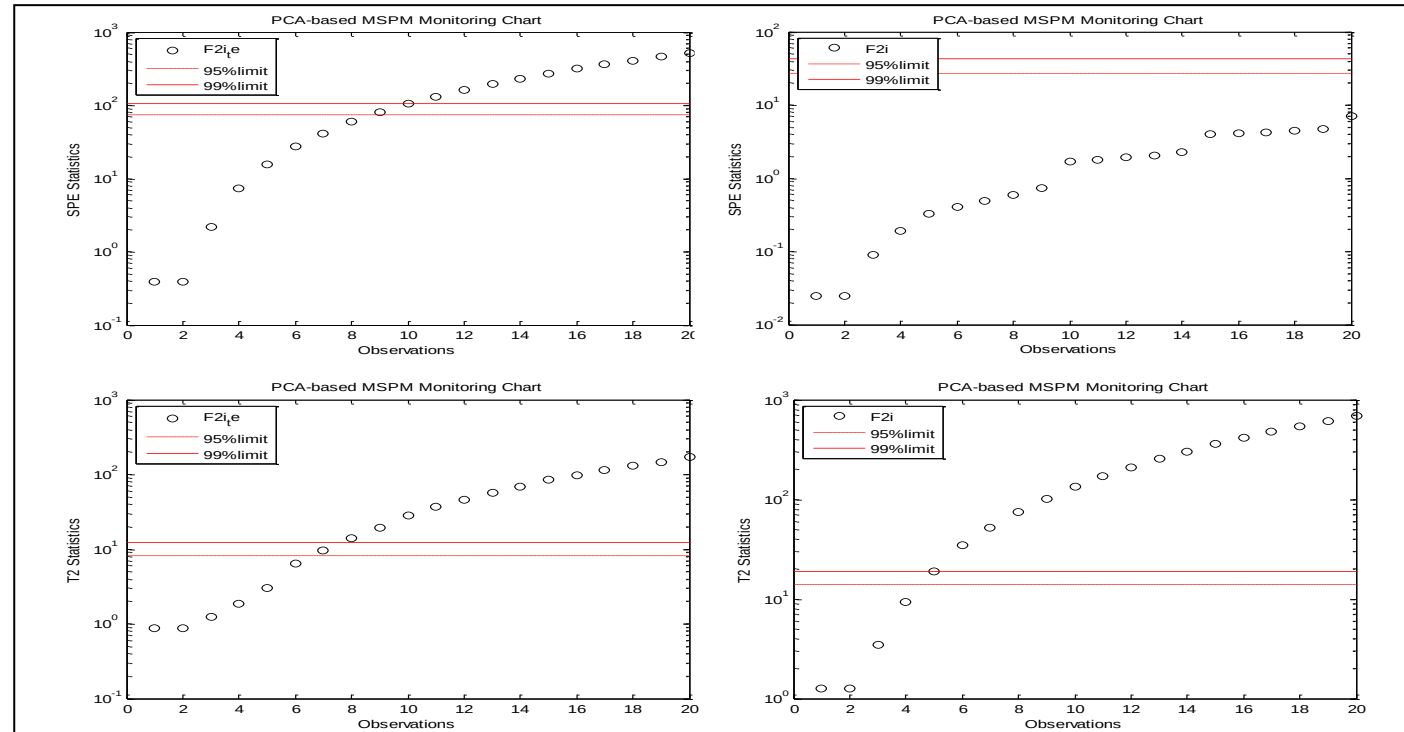
PHASE II: Fault Detection



Process Monitoring Overview

Motivation & Goal of Process Monitoring

PHASE II: Fault Detection



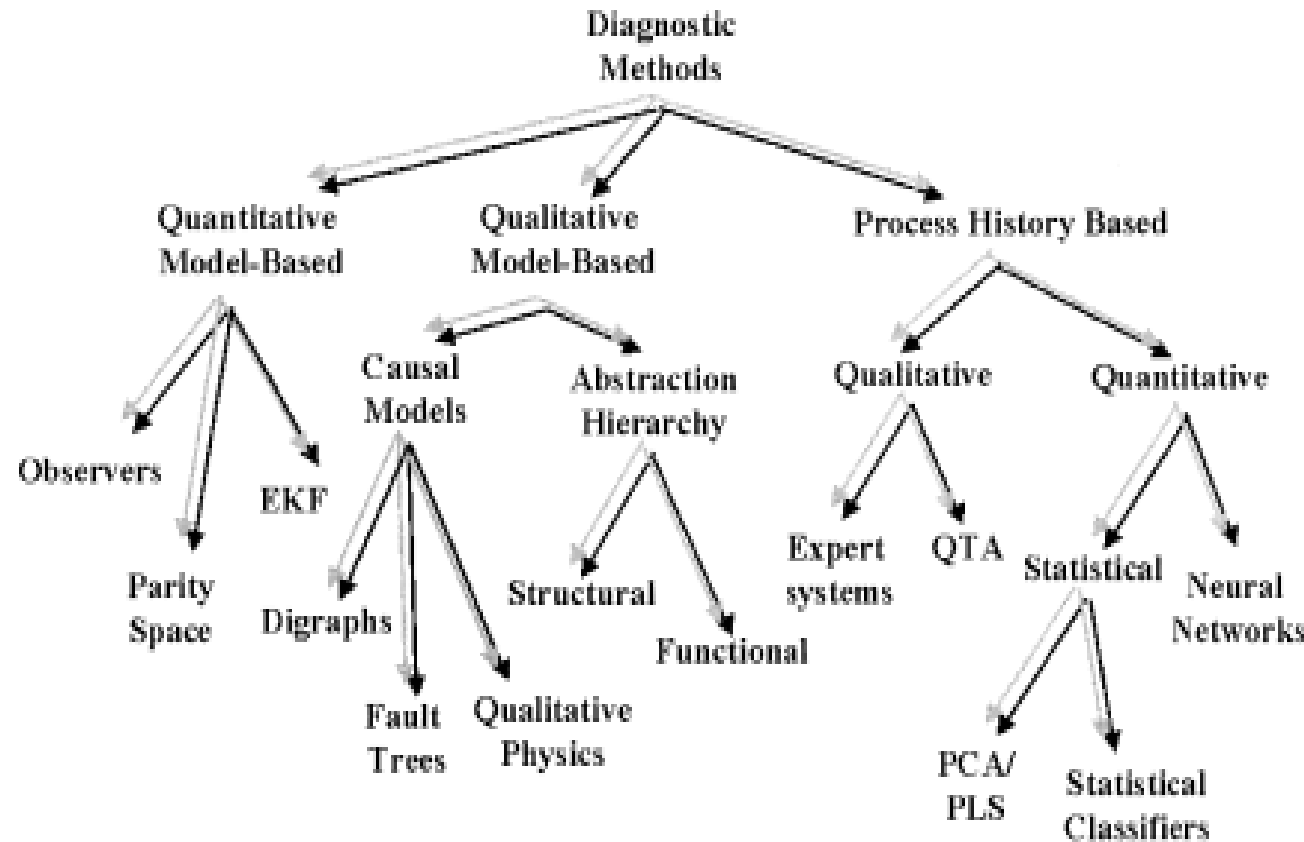
Process Monitoring Overview

Has your processing plant applied the monitoring system concept?



Process Monitoring Overview

Classification & Desirable Characteristics



Venkatasubramanian, V., Rengaswamy, R., Yin, K., Kavuri, S.N., (2003a). A Review of Process Fault Detection and Diagnosis. Part I: Quantitative model-based methods. *Computers and Chemical Engineering*, 27, 293 – 311.

Process Monitoring Overview

Classification & Desirable Characteristics

1. Quick Detection & Diagnosis
2. Isolability
3. Robustness
4. Novelty identification
5. Classification error estimation
6. Adaptability
7. Explanation facility
8. Modelling requirements
9. Storage and computational requirements
10. Multiple fault identification

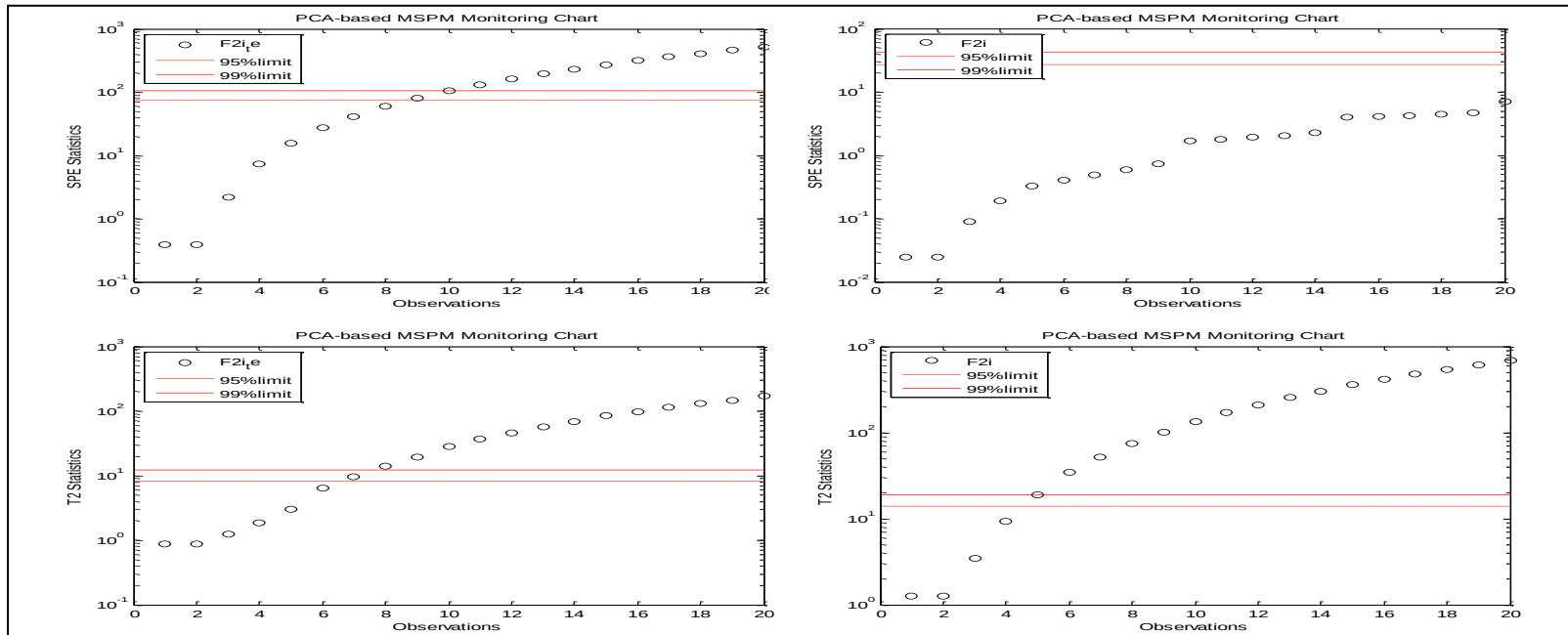
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Process Monitoring Overview

Classification & Desirable Characteristics

1. Quick Detection & Diagnosis

- The diagnostic system should respond quickly in detecting and diagnosing process malfunctions



Process Monitoring Overview

Classification & Desirable Characteristics

2. Iso-lability

- Iso-lability is the ability of the diagnostic system to distinguish between different types of failures

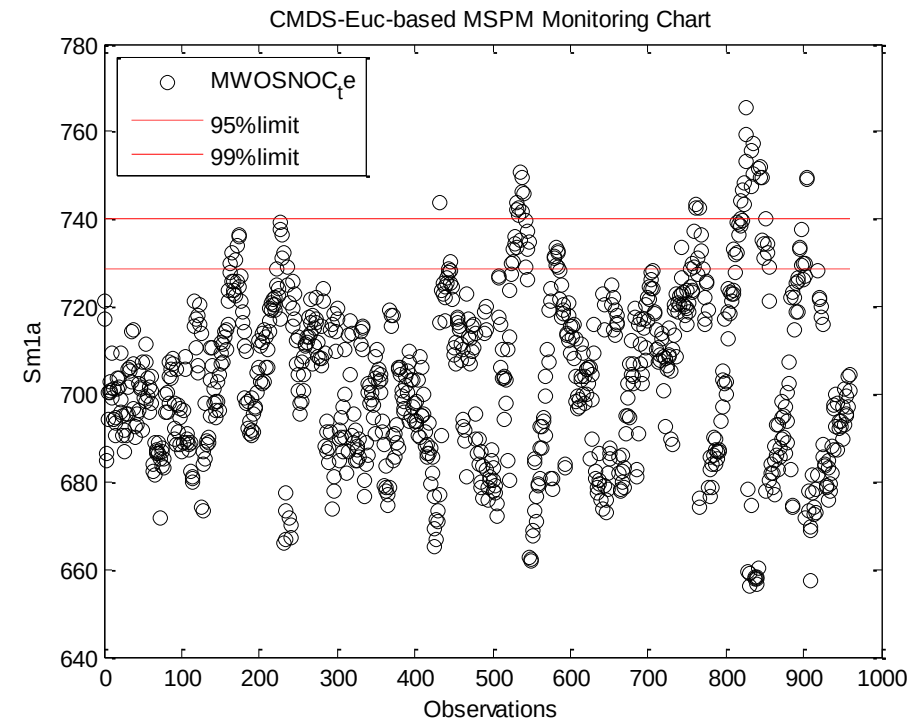
Fault Cases	Fault Causes
1	Pipe 1 blockage
2	External feed-reactant flow rate too high
3	Pipe 2 or 3 is blocked or pump fails
4	Pipe 10 or 11 is blocked or control valve 1 fails low
5	External feed-reactant temperature abnormal
6	Control valve 2 fails high
7	Pipe 7, 8, or 9 is blocked or control valve 2 fails low
8	Control valve 1 fails high
9	Pipe 4, 5, or 6 is blocked or control valve 3 fails low
10	Control valve 3 fails high
11	External feed-reactant concentration too low

Process Monitoring Overview

Classification & Desirable Characteristics

3. Robustness

- One would like the diagnostic system to be robust to various noise and uncertainties covering the whole mode of operations as much as possible.
- False alarm rate (FAR) analysis is conducted to evaluate the robustness of the monitoring statistics
- High FAR indicates that the developed limit settings are unsuitable to be used for monitoring

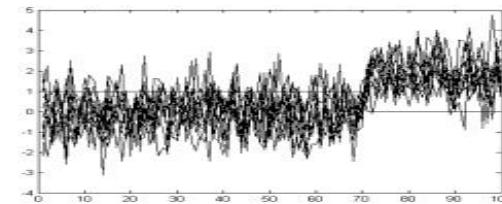


Process Monitoring Overview

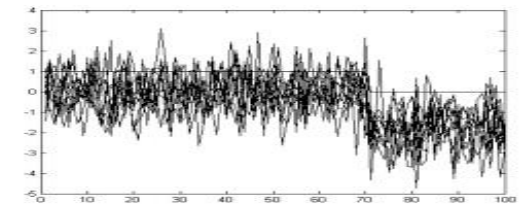
Classification & Desirable Characteristics

4. Novelty identification

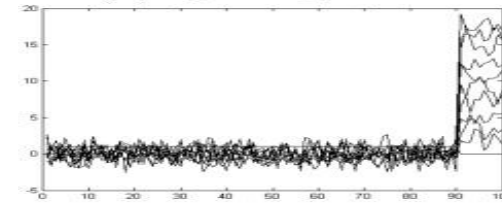
- One of the minimal requirements of a diagnostic system is to be able to decide, given current process conditions, whether the process is functioning normally or abnormally.
- If abnormal, whether the cause is a known malfunction or an unknown, novel malfunction.



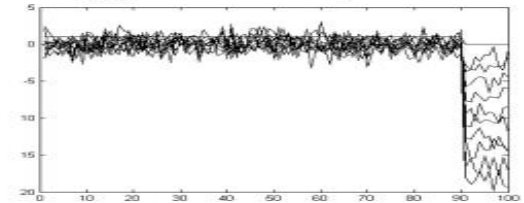
(a) Up shift pattern



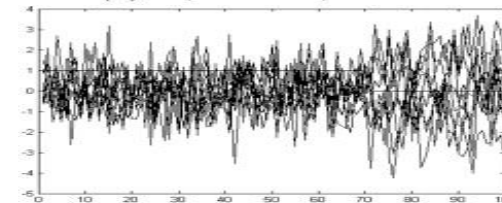
(b) Down shift pattern



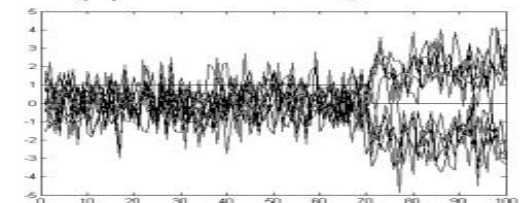
(c) Up trend pattern



(d) Down trend pattern



(e) Cyclic pattern

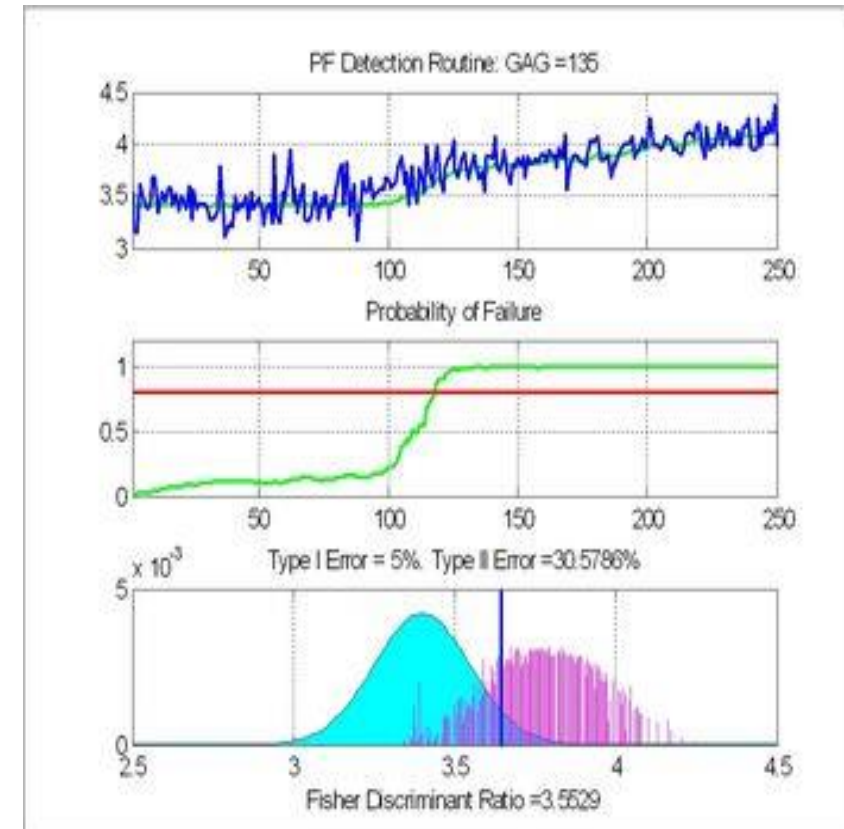


(f) Systematic pattern

Process Monitoring Overview

Classification & Desirable Characteristics

5. Classification error estimation
 - Such error measures would be useful to project confidence levels on the diagnostic decisions by the system giving the user a better feel for the reliability of the recommendations by the system.

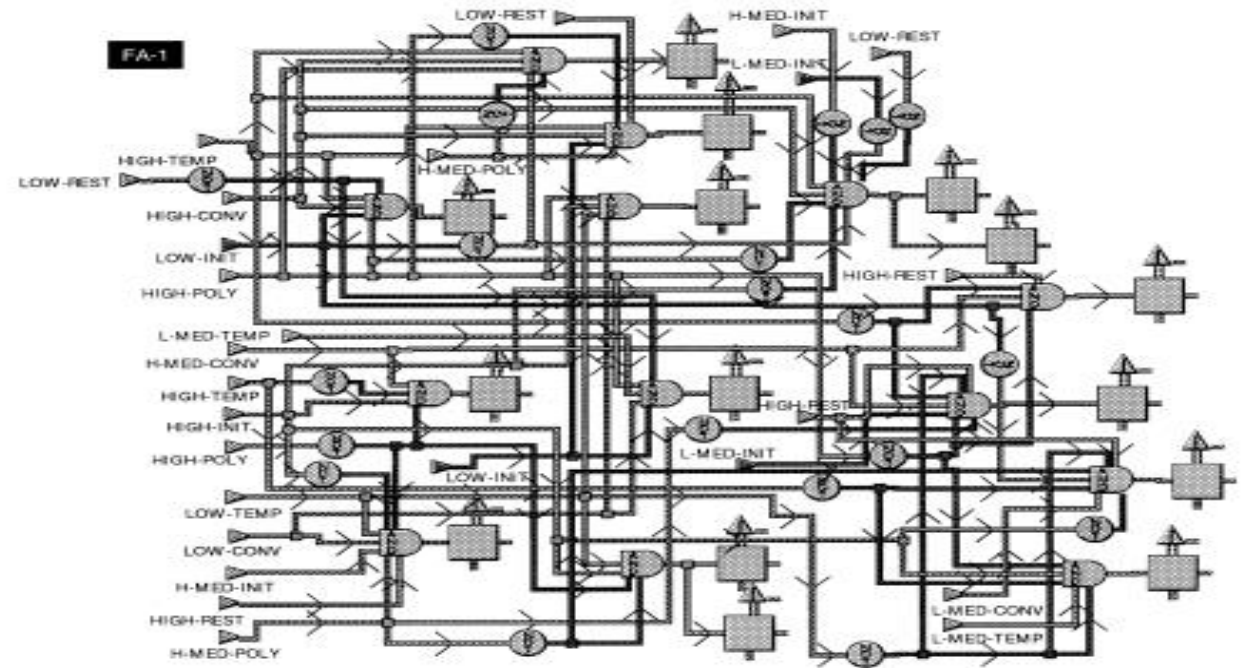


Process Monitoring Overview

Classification & Desirable Characteristics

6. Adaptability

- Processes in general change and evolve due to changes in external inputs or structural changes.
- Process operating conditions can change not only due to disturbances but also due to changing environmental conditions such as changes in production quantities with changing demands, changes in the quality of raw material etc.

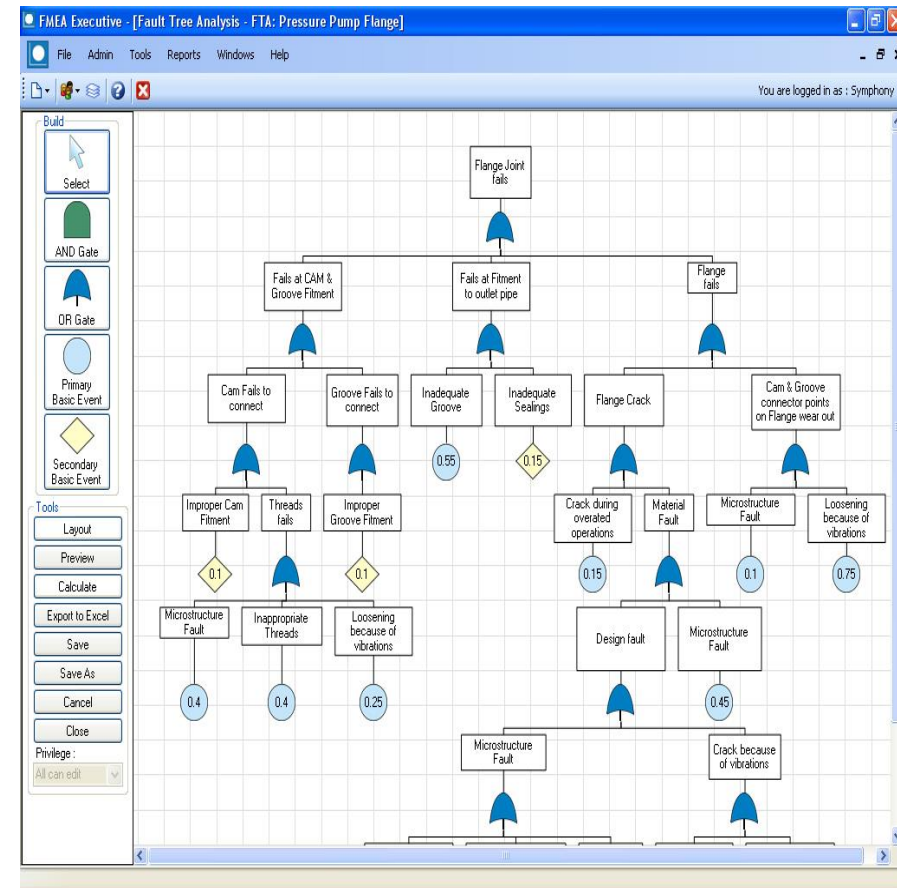


Process Monitoring Overview

Classification & Desirable Characteristics

7. Explanation facility

- Besides the ability to identify the source of malfunction, a diagnostic system should also provide explanations on how the fault originated and propagated to the current situation.
- This is a very important factor in designing on-line decision support systems.

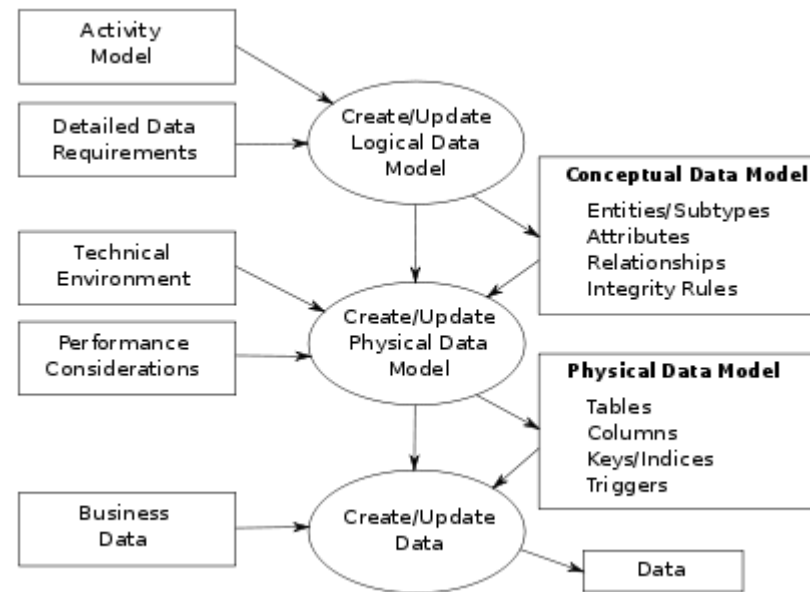


Process Monitoring Overview

Classification & Desirable Characteristics

8. Modelling requirements

- For fast and easy deployment of real-time diagnostic classifiers, the modelling effort should be as minimal as possible.

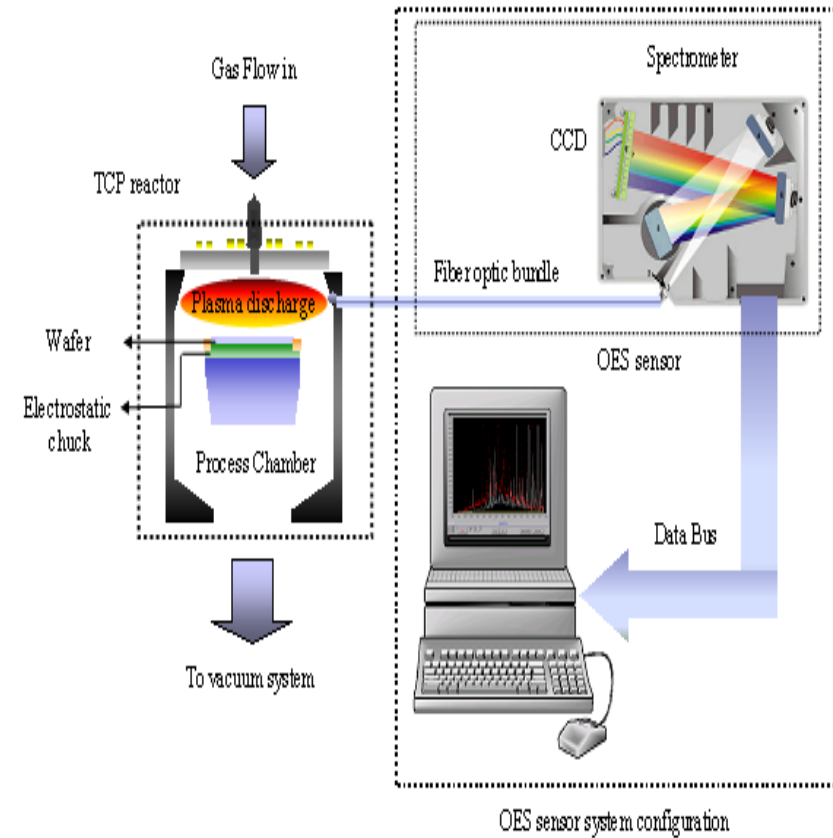


Process Monitoring Overview

Classification & Desirable Characteristics

9. Storage and computational requirements

- Usually, quick real-time solutions would require algorithms and implementations which are computationally less complex but might entail high storage requirements.
- One would prefer a diagnostic system that is able to achieve a reasonable balance on these two competing requirements.

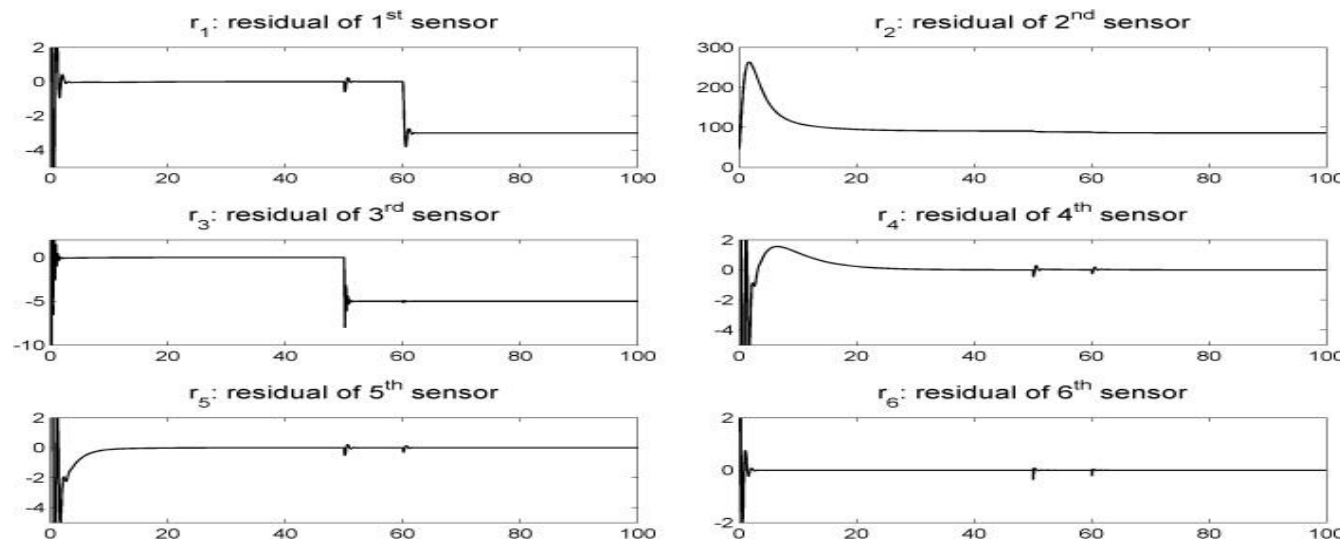


Process Monitoring Overview

Classification & Desirable Characteristics

10. Multiple fault identification

- The ability to identify multiple faults is an important but a difficult requirement. It is a difficult problem due to the interacting nature of most faults.



Process Monitoring Overview

Classification & Desirable Characteristics

	Observer	Digraphs	Abstraction hierarchy	Expert systems	QTA	PCA	Neural networks
Quick detection and diagnosis	✓	?	?	✓	✓	✓	✓
Isolability	✓	x	x	✓	✓	✓	✓
Robustness	✓	✓	✓	✓	✓	✓	✓
Novelty identifiability	?	✓	✓	x	?	✓	✓
Classification error	x	x	x	x	x	x	x
Adaptability	x	✓	✓	x	?	x	x
Explanation facility	x	✓	✓	✓	✓	x	x
Modelling requirement	?	✓	✓	✓	✓	✓	✓
Storage and computation	✓	?	?	✓	✓	✓	✓
Multiple fault identifiability	✓	✓	✓	x	x	x	x

Venkatasubramanian, V., Rengaswamy, R., Yin, K., Kavuri, S.N., (2003a). A Review of Process Fault Detection and Diagnosis. Part I: Quantitative model-based methods. *Computers and Chemical Engineering*, 27, 293 – 311.

Process Monitoring Overview

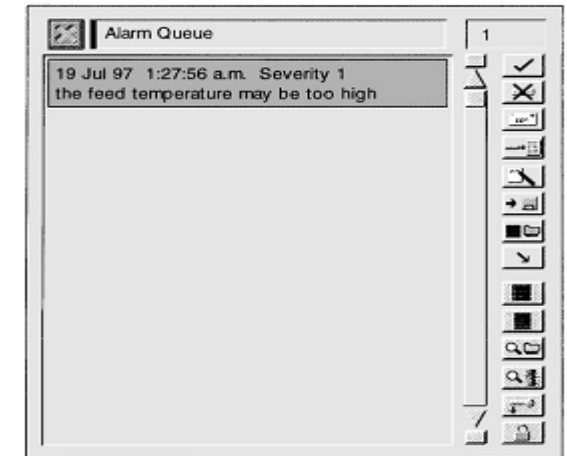
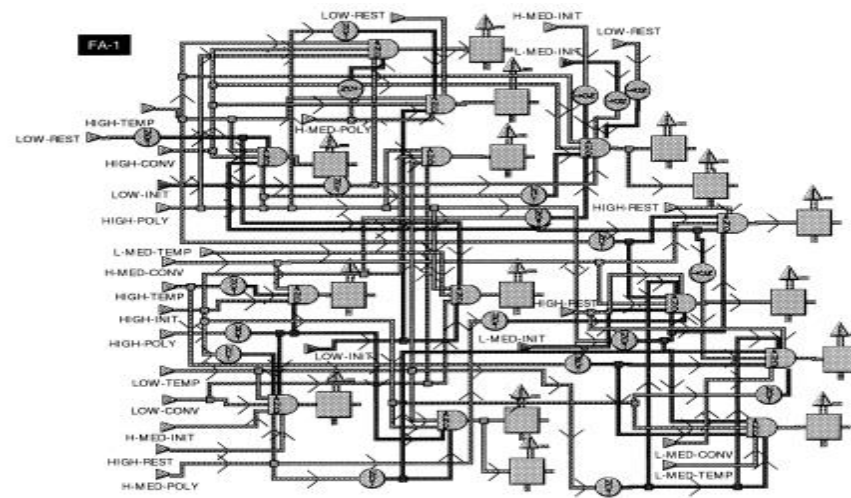
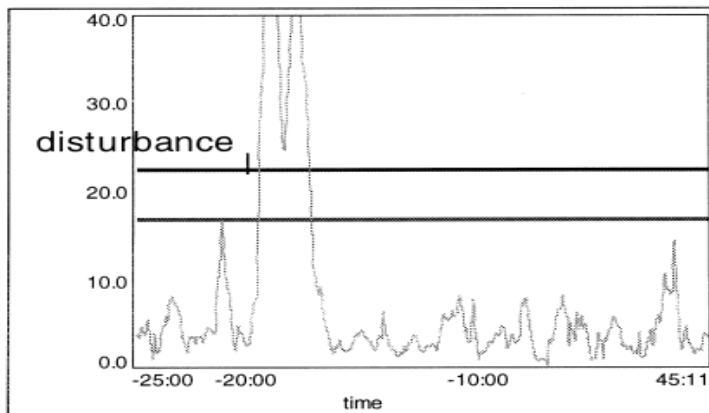
Classification & Desirable Characteristics

Intelligent process monitoring by interfacing knowledge-based systems and multivariate statistical monitoring

Aras Norvilas, Antoine Negiz¹, Jeffrey DeCicco, Ali Çinar*

Illinois Institute of Technology, Department of Chemical and Environmental Engineering, 10 W. 33rd Street, Chicago, Illinois, 60616, USA

T-Squared-Chart



Hybrid-Intelligent Monitoring System

Process Monitoring Overview

Which desirable characteristic that your plant has already implemented?

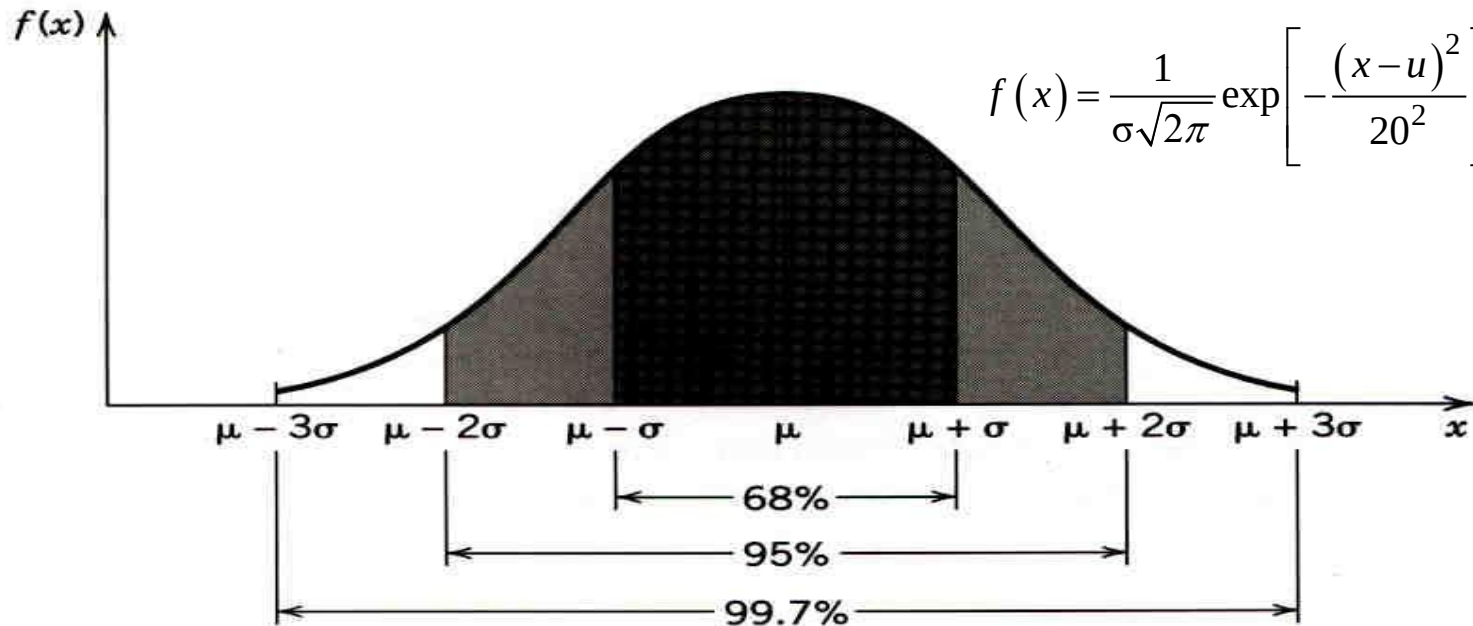


Exercise 1

- 1. Determine specifically what sort of faults/abnormal operations that your plant usually encountered in daily operation?**
- 2. How frequent those faults were usually happened?**

Statistical Process Control (SPC)

Normal Distribution Model



For the sake of generality, the tables are expressed in terms of the *standard normal distribution*, $N(0, 1)$, and the *standard normal variable*, $z = (x - \mu) / \sigma$.

Statistical Process Control (SPC)

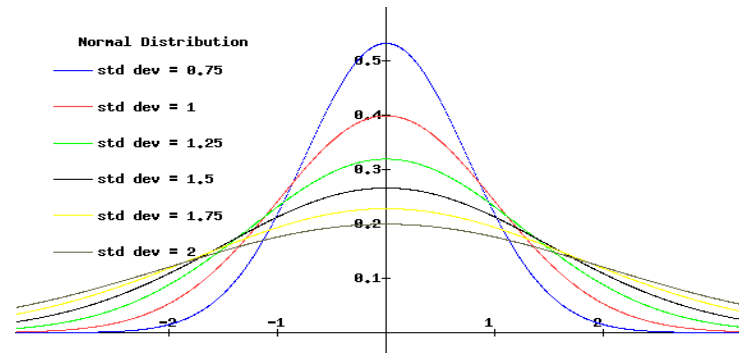
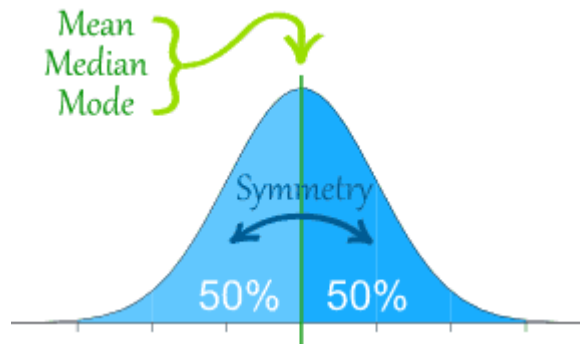
Normal Distribution Model

The aim of descriptive statistics: summarization of process data

Central tendency (magnitude of deviation from the targeted value)
Mode, median or **mean (average)**

Variability/dispersion
Range: Largest – smallest.
Sample variance

$$\bar{x}_i = \frac{\sum_{j=1}^n x_{j,i}}{n}$$



$$\sigma^2_i = \frac{1}{n-1} \sum_{j=1}^n (x_{j,i} - \bar{x}_i)^2$$

Statistical Process Control (SPC)

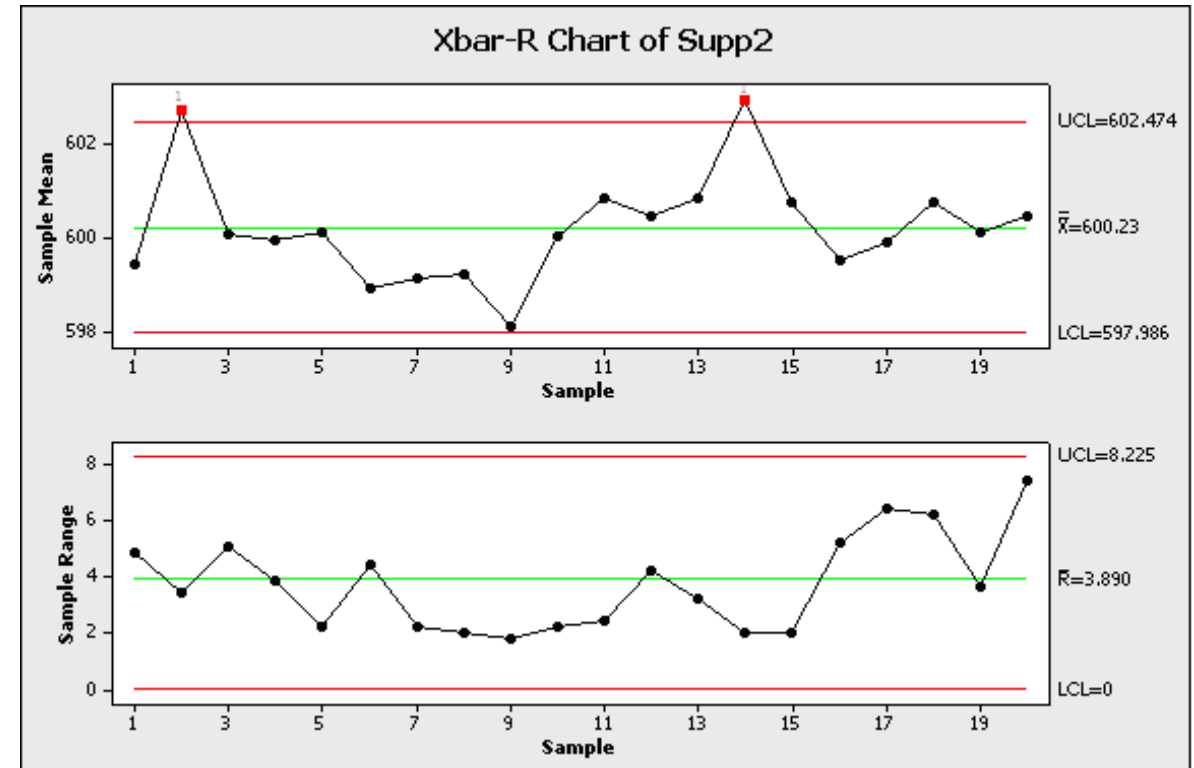
Development of Control Charts

- In statistical process control, *Control Charts* (or *Quality Control Charts*) are used to determine whether the process operation is normal or otherwise.
- This type of control chart is often referred to as a Shewhart Chart, in honor of the pioneering statistician, Walter Shewhart, who first developed it in the 1920s.

Statistical Process Control (SPC)

Development of Control Charts

- ☑ For variables that have continuous dimensions
- ☑ Weight, speed, length, strength, etc.
- ☑ x-charts are to control the central tendency of the process
- ☑ R-charts are to control the dispersion of the process
- ☑ These two charts must be used together



Statistical Process Control (SPC)

Development of Control Charts

1. Take representative sample from output of a process over a long period of time, e.g. 10 units every hour (I) for 24 hours (n).
2. Compute means and ranges for the variables and calculate the control limits
3. Draw control limits on the control chart
4. Plot a chart for the means and another for the mean of ranges on the control chart
5. Determine state of process (in or out of control)
6. Investigate possible reasons for out of control events and take corrective action
7. Continue sampling of process output and reset the control limits when necessary.

Statistical Process Control (SPC)

Development of Control Charts

For $\bar{x}$ -Charts when we know σ

$$\text{Upper control limit (UCL)} = \bar{\bar{x}} + z\sigma_x$$

$$\text{Lower control limit (LCL)} = \bar{\bar{x}} - z\sigma_x$$

where

- $\bar{\bar{x}}$ = mean of the sample means or a target value set for the process
- z = number of normal standard deviations
- σ_x = standard deviation of the sample means
= $\sigma/\sqrt{l}$
- σ = population standard deviation
- l = sub-group size

Where typically, $z = 3$.

Statistical Process Control (SPC)

Development of Control Charts

Hour 1			
Observation Number	Weight of Oat Flakes	Hour	Mean
I = 9 {	1	17	
	2	13	
	3	16	
	4	18	
	5	17	
	6	16	
	7	15	
	8	17	
	9	16	
Mean 16.1			
Given: $\sigma = 1$			

Hour	Mean	Hour	Mean
1	16.1	7	15.2
2	16.8	8	16.4
3	15.5	9	16.3
4	16.5	10	14.8
5	16.5	11	14.2
6	16.4	12	17.3

For 99.73% control limits, $z = 3$

$$UCL_x = \bar{\bar{X}} + z\sigma_x = 16 + 3(1/3) = 17 \text{ ozs}$$

$$LCL_x = \bar{\bar{X}} - z\sigma_x = 16 - 3(1/3) = 15 \text{ ozs}$$

Statistical Process Control (SPC)

Development of Control Charts

For R-Charts

$$\text{Upper control limit (UCL}_R\text{)} = D_4 \bar{R}$$

$$\text{Lower control limit (LCL}_R\text{)} = D_3 \bar{R}$$

where

$\bar{R}$ = average range of the samples

D_3 and D_4 = control chart factors from Table 1

Subgroup Size I	Mean Factor A_2	Upper Range D_4	Lower Range D_3
2	1.880	3.268	0
3	1.023	2.574	0
4	.729	2.282	0
5	.577	2.115	0
6	.483	2.004	0
7	.419	1.924	0.076
8	.373	1.864	0.136
9	.337	1.816	0.184
10	.308	1.777	0.223
12	.266	1.716	0.284

Statistical Process Control (SPC)

Development of Control Charts

Average range $\bar{R} = 5.3$ pounds

Sample size $l = 5$

From Table S1 $D_4 = 2.115$, $D_3 = 0$

$$\begin{aligned} UCL_R &= D_4 \bar{R} \\ &= (2.115)(5.3) \\ &= 11.2 \text{ pounds} \end{aligned}$$

$$\begin{aligned} LCL_R &= D_3 \bar{R} \\ &= (0)(5.3) \\ &= 0 \text{ pounds} \end{aligned}$$

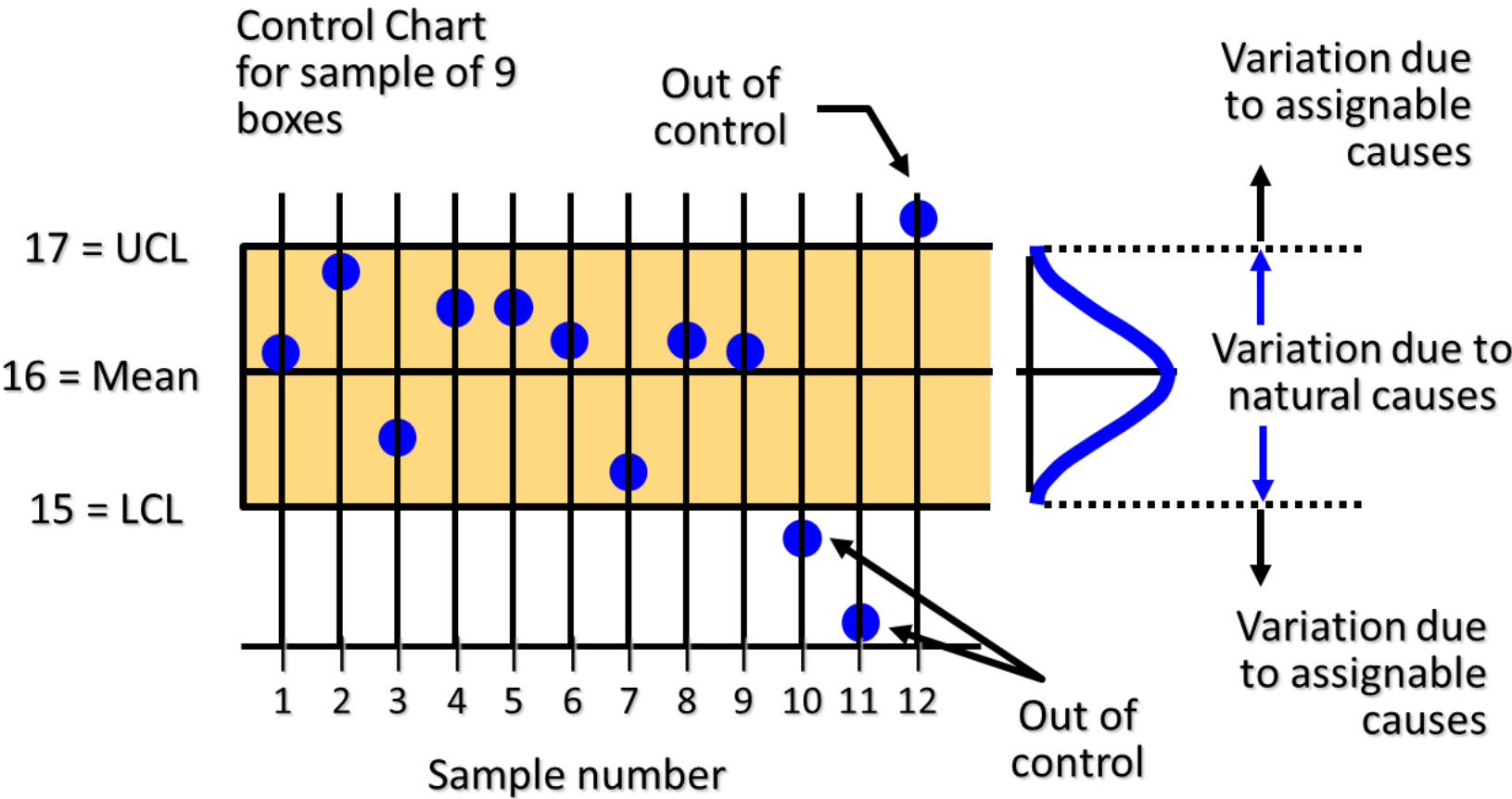
UCL = 11.2

Mean = 5.3

LCL = 0

Statistical Process Control (SPC)

Interpretation of Control Charts

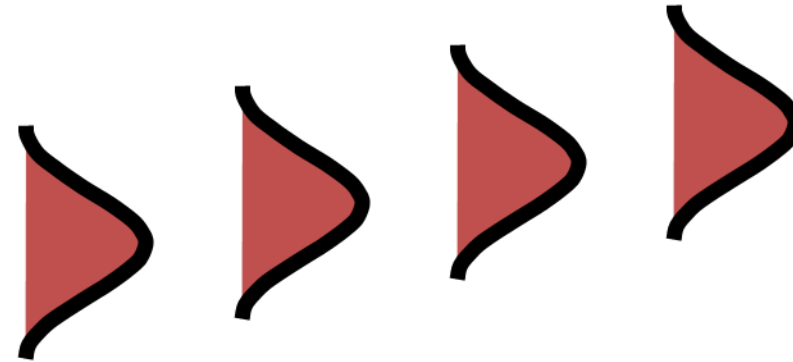


Statistical Process Control (SPC)

Interpretation of Control Charts

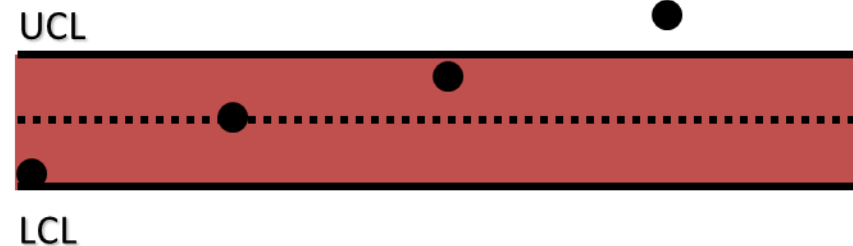
(a)

These sampling distributions result in the charts below



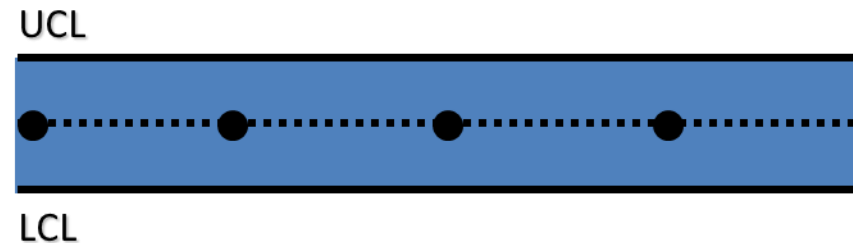
(Sampling mean is shifting upward but range is consistent)

$\bar{x}$ -chart



($\bar{x}$ -chart detects shift in central tendency)

R-chart



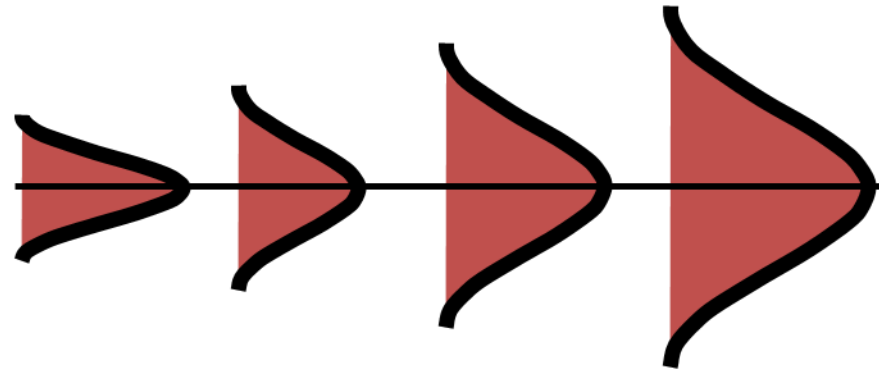
(R-chart does not detect change in mean)

Statistical Process Control (SPC)

Interpretation of Control Charts

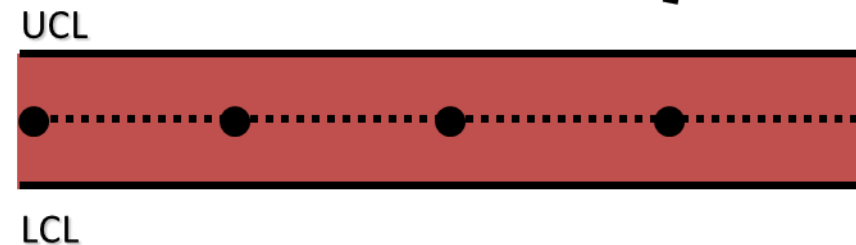
(b)

These sampling distributions result in the charts below



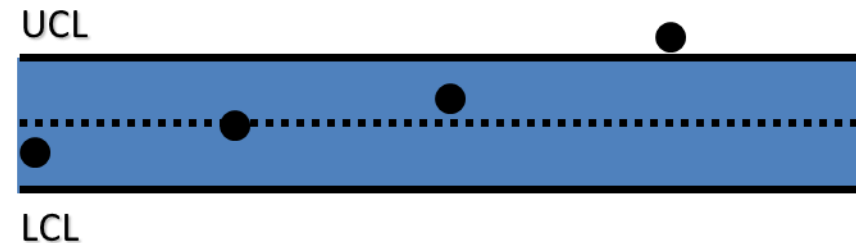
(Sampling mean is constant but dispersion is increasing)

$\bar{x}$ -chart



($\bar{x}$ -chart does not detect the increase in dispersion)

R-chart



(R-chart detects increase in dispersion)

Statistical Process Control (SPC)

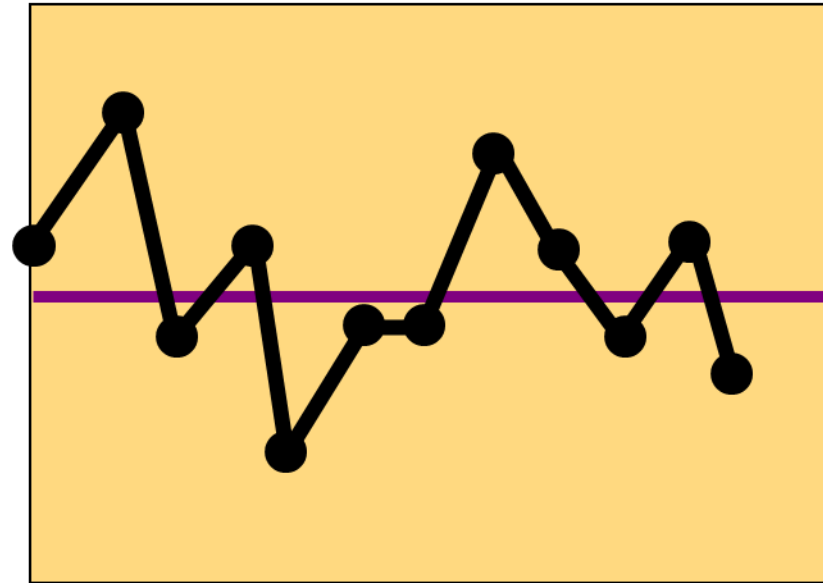
Interpretation of Control Charts

Patterns

Upper control limit

Target

Lower control limit



Normal behavior. Process is “in control.”

Statistical Process Control (SPC)

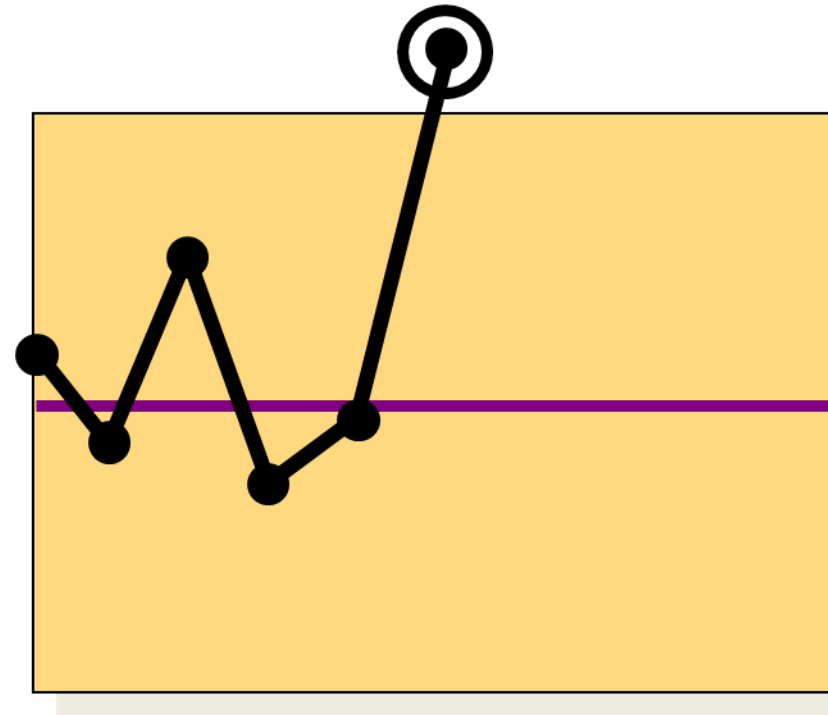
Interpretation of Control Charts

Patterns

Upper control limit

Target

Lower control limit



One plot out above (or below).
Investigate for cause. Process is not
necessary “out of control.”

Statistical Process Control (SPC)

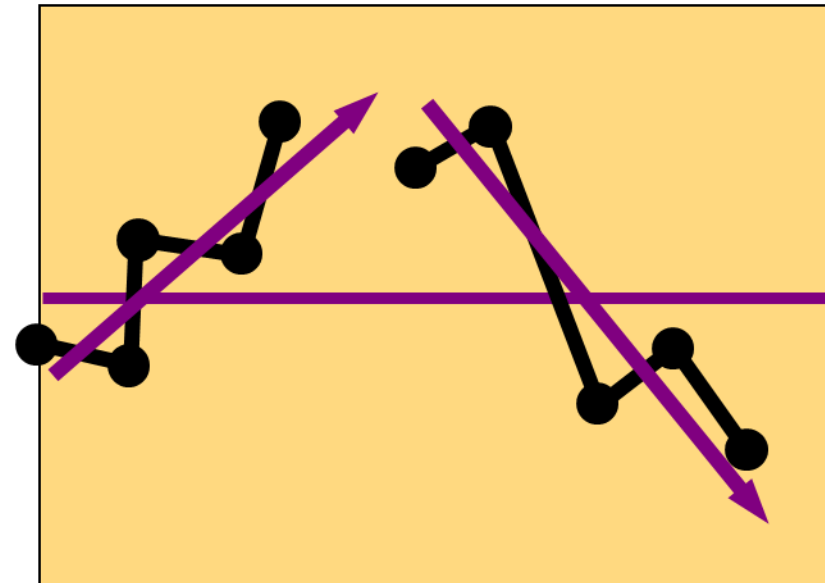
Interpretation of Control Charts

Patterns

Upper control limit

Target

Lower control limit



Trends in either direction, 5 plots.
Investigate for cause of progressive
change.

Statistical Process Control (SPC)

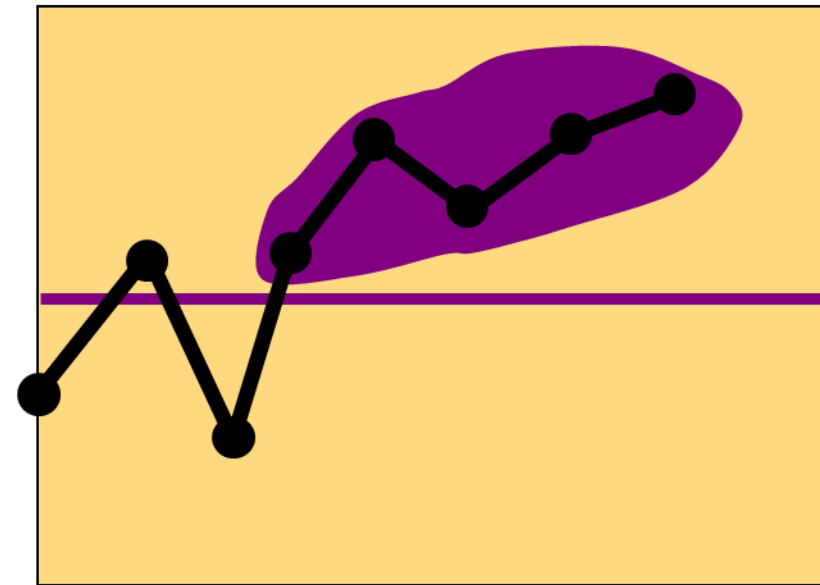
Interpretation of Control Charts

Patterns

Upper control limit

Target

Lower control limit



Run of 5 above (or below) central line. Investigate for cause.

Exercise 2

1. **Develop the SPC control charts based on the specified NOC & CASE data of CSTR system that provided.**
2. **Develop and determine the appropriateness of using the SPC charts for process enhancement in the BPC plant operation.**